

Register Number

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**Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110****(An Autonomous Institution, Affiliated to Anna University, Chennai)****Computer Science and Engineering****Continuous Assessment Test – I****Regulations – R2021****Question Paper**

Degree & Branch	B.E - CSE				Semester	V	
Subject Code & Name	UCS2503 SOFTWARE ENGINEERING						
Academic Year	2023-2024 ODD	Batch	2021-2025	Date	22.09.2023	FN	
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks		

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1:	Choose an appropriate process model and estimate project cost and effort required by applying software engineering principles(K5)
CO2:	Identify and analyze the requirements and construct their models(K5)
CO3:	Apply systematic procedure for software design(K3)
CO4:	Compare and contrast the various testing and maintenance activities (K2)
CO5:	Make use of agile development and Devops (K3)
CO6:	Identify unethical issues and apply ethical practices for a given case study(K3)

Part – A (4 × 2 = 8Marks)

		KL	CO	PI
1.	What are the best practices for successful software development ?	K1	CO1	1.3.1
2.	Which phase in the SDLC is the most important ? Identify the reasons.	K3	CO1	2.1.2
3.	What is feasibility in project planning?	K1	CO2	1.4.1
4.	Give some examples of usability requirement for IRCTC website.	K2	CO2	1.4.1 2.1.2

Part – B (3×6 = 18 Marks)

		KL	CO	PI
5.	Differentiate between prototyping model and spiral model.	K2	CO1	1.3.1 1.4.1
6.	Identify and list any three user requirements, and any three system requirements in Moodle LMS software.	K3	CO2	3.1.2
7.	Explain cohesion and coupling in software engineering. Give examples.	K2	CO3	3.2.2

Part – C (2 × 12 = 24 Marks)
(Any subdivisions should be either 7+5 or 8+4)

		KL	CO	PI																											
8.	a. Differentiate LOC and function point estimation. (5) b. Given the function point data (7) i. A target product has 8 simple inputs, 4 average inputs, and 12 complex inputs. ii. There are 66 average outputs, 18 simple inquiries, 10 average master files, and 7 complex interfaces. iii. Consider the total degree of influence for the product is 46 <i>(46)</i>. Estimate the number of function points.	K3	CO1	2.1.2 1.3.1																											
Table 1: Function point evaluation <table><tr><th rowspan="2">Component</th><th colspan="3">Level of Complexity</th></tr><tr><th>Simple</th><th>Average</th><th>Complex</th></tr><tr><td>External Inputs</td><td>3</td><td>4</td><td>6</td></tr><tr><td>External outputs</td><td>4</td><td>5</td><td>7</td></tr><tr><td>External Inquiries</td><td>3</td><td>4</td><td>6</td></tr><tr><td>Internal logical files</td><td>7</td><td>10</td><td>15</td></tr><tr><td>External interface files</td><td>5</td><td>7</td><td>10</td></tr></table>					Component	Level of Complexity			Simple	Average	Complex	External Inputs	3	4	6	External outputs	4	5	7	External Inquiries	3	4	6	Internal logical files	7	10	15	External interface files	5	7	10
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(Or)																															
9.	Design the network diagram for a project associated with the activities given in the table below for ABC consulting company project. (5) How long should it take for project manager and his team to complete this project? Identify the critical path activities. (7)	K3	CO1	1.3.1 2.1.2																											
<table><tr><th>Activity</th><th>Immediate Predecessor(s)</th><th>Time (Days)</th></tr><tr><td>A</td><td>-</td><td>3</td></tr><tr><td>B</td><td>A</td><td>4</td></tr><tr><td>C</td><td>A</td><td>6</td></tr><tr><td>D</td><td>B</td><td>6</td></tr><tr><td>E</td><td>B</td><td>4</td></tr><tr><td>F</td><td>C</td><td>4</td></tr><tr><td>G</td><td>D</td><td>6</td></tr><tr><td>H</td><td>E, F</td><td>8</td></tr></table>					Activity	Immediate Predecessor(s)	Time (Days)	A	-	3	B	A	4	C	A	6	D	B	6	E	B	4	F	C	4	G	D	6	H	E, F	8
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H	E, F	8																													
10.	a) List the non-functional requirements for the following case study. Consider an automated library circulation system. Every book has a bar code, and every borrower has a card bearing a bar code. When a borrower wishes to check out a book, the librarian scans the bar codes on the book and the borrower's card, and enters C at the computer terminal. Similarly, when a book is returned, it is	K3	CO2	3.1.2 1.4.1 2.1.2 2.3.2																											

	again scanned and the librarian enters R. Librarians can add books (+) to the library collection or remove them (-). Borrowers can go to a terminal and determine all the books in the library by a particular author (the borrower enters A= followed by the author's name), all the books with a specific title (T= followed by the title), or all the books in a particular subject area (S= followed by the subject area). Finally, if a borrower wants a book currently checked out, the librarian can place a hold on the book so that, when it is returned, it will be held for the borrower who requested it (H= followed by the number of the book). (5) b) What is requirement elicitation? Apply elicitation techniques to gather requirements on a scenario to include the friend list in a whatsapp software. (7)			
(Or)				
11.	For the case study described in question no. 10 (a) Draw the level – 0 DFD, level-1 DFD diagram. (5+7)	K3	CO2	3.1.2 1.4.1 2.1.2 2.3.2